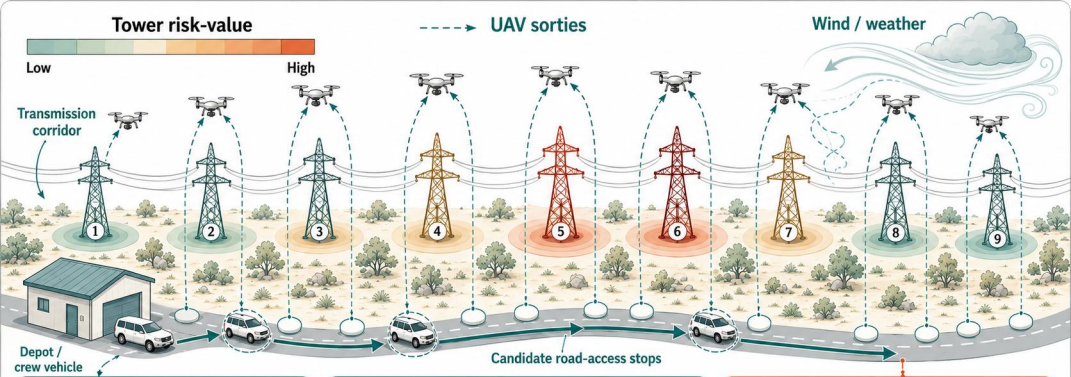




q95 energy screening



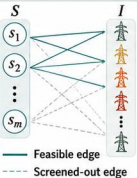
$$Q_{si}^{0.95} = \mu_{si} + z_e \sigma_{si}$$

$$Q_{si}^{0.95} \leq B_d(1-\rho)$$

$$Q_{si}^{0.95} > B_d(1-\rho) \Rightarrow y_{si} = 0$$

$$y_{si} \leq a_s, \quad y_{si} \in \{0, 1\}$$

Feasible service graph



$$\mathcal{F} = \{(s, i) \in S \times I : Q_{si}^{0.95} \leq B_d(1-\rho)\}$$

$$\sum_{s: (s, i) \in \mathcal{F}} y_{si} = 1, \quad y_{si} \leq a_s$$

$$y_{si} \in \{0, 1\}$$

Risk-value schedule

Early



Later

$$\begin{aligned} \min \quad & \lambda_1 \sum_{k \in K} \sum_{(u,v) \in A} t_{uv}^k x_{uv}^k \\ & + \lambda_2 \sum_{(s,i) \in \mathcal{F}} Q_{si}^{0.95} y_{si} \\ & + \lambda_3 \sum_{i \in I} C_i + \lambda_4 \sum_{i \in I} r_i v_i C_i \end{aligned}$$